# SCHOOL OF ADVANCED SCIENCES (SAS)

## About School

The School of Advanced Sciences (SAS) was established in 2017 at VIT-AP University. Currently, Mathematics, Physics and Chemistry departments constitute the school. We have about 121 excellent faculty backed by a wealth of research and teaching experience from Institutes and Universities of International repute. We have excellent infrastructure for teaching and research in conjunction with a digital library and advanced computational facilities. The teaching and learning process at VIT-AP is carefully crafted and mellifluously executed with the moto of "Apply Knowledge. Improve life!" to transform students into a work force of competent and responsible global citizens. SAS offers a number of foundational and advanced level courses to 300 BSc and MSc students with in and to the other schools of the University. We have 200 research scholars in SAS working in diverse areas of science and technology towards their PhDs. We believe strongly that great accomplishments are possible when diverse ideas merge through collaborations. Our faculty collaborate extensively with industry and academia for sustainable solutions to global problems.

## Eligibility

- Minimum of 55% marks at 10+2 level / Intermediate / CBSE / ICSE / HSC / or equivalent with Mathematics as one of the compulsory subjects.
- 10% waiver is applicable for the students belonging to SC/ST category and Jammu and Kashmir / Ladakh & North-eastern states except the Law Programme
- Undergraduate degrees such as BA., B.Sc., B.Math., B.Stat., B.E., B.Tech., etc. with 60% marks, with a background in Mathematics or Statistics or Computer Science or Data Science
- B.Sc Degree from any recognized university with Physics / Applied Science with 60% of marks.
- B.Sc degree in either Chemistry or Applied Chemistry or Applied Science from any recognized university with Physics or Mathematics or Botany or Zoology as ancillary subjects with 60% of marks
- A Master's degree in the relevant discipline of science with a minimum of 60% marks or a CGPA of 6.5 on a 10-point scale is mandatory.

## Age Limit

- Candidates whose date of birth falls on or after 1st July 2005 are eligible to apply for UG admission 2025

# DUAL DEGREE

## B-SC-M-SC-DATA-SCIENSE

## About the Programme

This programme encompasses areas such as developing accurate and efficient numerical methods for solving physical or biological models, analysis of numerical approximations to differential and integral equations, developing computational tools to better understand data and structure, etc. Many industry experts and popular business publications have attested that 'data' in the 21st Century is, like 'oil' in the 18th Century: the most valuable resource in the world. Hence, the smooth functionality of everything from the government to local companies depend on how we understand data, without which the progress would halt. For those who realize the fundamental value of data and learn to extract and use it, there will be huge rewards in the current data-driven economy. Therefore, for those interested in Mathematics, Statistics, and Computer Science, this programme offers promising career opportunities with high-salaried packages.

## Key Features

1. SAS inks Academic & Research MoU with CSIR-IICT
2. SAS inks Industrial MoU with Mitsui Kinzoku Components India Private Limited (MKCI)
3. SAS inks Academic & Research MoU with Slovak University of Agriculture in Nitra, Slovakia, Europe
4. Ongoing more than 7 External Funded Projects by DST, SERB, ICMR, UGC-DAE, etc.
5. 14 Indian Patents publications by faculty and students of SAS
6. 595 research publications by faculty and students of SAS
7. UG/PG students are working on many innovative projects/research
8. Good placements for the graduated students
9. Reputed faculty members with Ph.D./Postdoctoral qualification
10. Exclusive MoU offering a 2.5+1.5 program for Data Science students with Binghamton University
11. The curriculum is according to NEP-2020

## Career Prospects

According to Harvard Business Review, "Data Scientist is a high-ranking professional with the expertise and curiosity to make discoveries in the world of Big Data". As the world is increasingly turning to data for decision making, and the amount of data being generated is growing exponentially, there is huge demand for Data Scientists in start-ups as well as well-established companies as they rush to hire Data Scientists to tame their business-critical Big Data. In any organization, a data scientist does model-driven analysis of its data to help it to improve planning and increase productivity. Data scientists are needed to work at various levels of business decision-making, such as tactical, operational, and strategic levels, and to share insights with the business. Therefore, Data Scientists are coveted professionals in industries and sectors such as IT industries, research institutions, Banking and Financial sector, educational sector, consulting firms, pharmaceutical companies, automobile industries and so on. Furthermore, the students are also absorbed in research-oriented government and private industries, doctoral programmes in leading universities, as well as software-based companies.

## Program Outcomes

After successful completion of the program a student is expected to have abilities to

**Disciplinary Knowledge:**

Apply the knowledge of mathematics, Statistics and Computer Science fundamentals, and a Data Science specialization to the solution of complex data-driven problems.

**Problem analysis:**

Identify, formulate, research literature, and exhibit critical thinking skills to analyse complex data driven problems, reaching substantiated conclusions using the fundamental principles of mathematics, statistics, computing and data sciences

**Design/Development of solutions/Problem Solving:**

Design solutions for data-driven problems and apply lateral thinking and analytical skills to design systems or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

**Conduct investigations of complex problems:**

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

**Modern tool usage:**

Create, select, and apply appropriate techniques, resources, and computing tools including prediction and modelling to complex decision-making activities related to data with an understanding of the limitations.

**The Data Science and society:**

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

**Environment and sustainability:**

Understand the impact of the professional data science solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**Ethics:**

Demonstrate empathetic social concern; equity centred national development; apply ethical principles; commit to professional ethics & responsibilities and norms of the data science practice.

**Individual and teamwork:**

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**Communication:**

Communicate effectively on complex data science activities with the scientific community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

**Project management and Finance:**

Demonstrate knowledge and understanding of the scientific, engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

**Life-long learning:**

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of socio-technological change.

## Programme Educational Objectives

- Apply Data Science theory blended with Mathematics, Statistics, and Computer Science to model and solve data-driven problems to take decisions in the best interest of the organizations and society.
- Practice value-based holistic and comprehensive learning by integrating traditional and innovative learning practices to match the highest quality standards and groom the students to be effective leaders in their chosen fields and career.
- Unleash the hidden talents and creative potential of the students and nurture their spirit of critical thinking and to pursue higher education.
- Enable them to adapt themselves better to the changing global scenario and gain access to versatile career opportunities in multidisciplinary domains.
- Develop a sense of social responsibility, ethics, and equity to transform themselves to commitment-oriented professionals having a strong attitude towards sustainable development of the nation.

## Programme Specific Outcome

- Demonstrate an understanding of fundamental concepts in Mathematics, Statistics, and Computer Science and usage of various data analysis tools and computing software used in Data Science problem solving.
- Gain personal and professional competence to be able to collect, process data, model, analyse, design, visualize to address real-world problems, and interpret results using Data Analytics.
- Acquire capabilities to execute research experiments and projects; to design and develop innovative processes, tools, and algorithms; and to build predictive models and machine learning algorithms to analyse large amounts of information from structured and unstructured data to discover trends and patterns.
- Identify and analyse social, legal, and ethical issues in Data Science; provide solutions to particular domain specific problems by having in depth domain knowledge; get exposure to emerging trends and technologies to prepare students for industry and higher studies.

# POSTGRADUATE

## M-SC-PHYSICS

### About the Programme

The M.Sc. Physics programme aims to develop an application and technology oriented learning, along with an in-depth training of the core classical areas of Physics. The programme is designed with a combination of theoretical, experimental, and project-based learning components of Physics to enhance the passion for the subject, a deep understanding on various interdisciplinary areas, and a desire to contribute for the development of new technologies. Under this programme, students get an opportunity to get introduced to cutting edge modern day technologies and gain experience via active hands-on learning. The programme is of two years duration and is split into four semesters. Students have the opportunity to get engaged in interim and final semester projects, and work in the many interdisciplinary areas of research active in the University. You may please contact our faculty members for further details regarding the curriculum.

## Key Features

1. SAS signed Academic & Research MoU with CSIR-IICT

2. SAS signed Industrial MoU with Mitsui Kinzoku Components India Private Limited (MKCI)

3. Department of Physics has MoU with NoPo Nanotechnologies and QpiAI Private Limited, India

4. Department of Physics has four ongoing DST-funded projects and five DAE-UGC- CRS projects worth nearly 1.5 core.

5. 15 Indian Patent publications by faculty and students of physics

6. 150 research publications by faculty and students of physics

7. Well experienced doctoral faculty with an excellent track record of publications and patents

## Career Prospects

M.Sc. Physics graduates have promising career opportunities in the academia and industry. This includes PhD opportunities in reputed national and international institutes and universities; Research Scientist positions in Government Labs and Industrial R&Ds; and teaching and research positions in academia. The job sector includes a wide variety of positions in Instrumentation, Materials Development, Manufacturing in rapidly developing Electric Vehicle Industries, Semiconductor fabrication, Defence, Energy, Automotive, Medical, Aerospace, Robotics, and AI.

## Program Outcomes

After successful completion of the program a student is expected to have abilities to

**Disciplinary Knowledge**

Acquire knowledge of fundamentals of general Physics, along with specialization areas offered and contemporary issues

**Problem analysis**

Solve analytical, practical, and research-based problems, reaching substantiated conclusions using the fundamental principles of physics

**Design/Development of solutions/Problem Solving**

Develop concept-based problem-solving skills, through critical thinking to approach physical problems, both in theory and experiments

**Conduct investigations of complex problems**

apply conceptual knowledge and advanced methods for analysis and synthesis of the information to provide valid conclusions

**Modern tool usage**

Apply appropriate techniques, resources, experimental and computational tools related to physical sciences with an understanding of the limitations

**Physics and society**

Develop rational and scientific thinking based on the contextual knowledge to appreciate societal issues in general

**Environment and sustainability**

Understand the impact of the professional scientific solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

**Ethics**

Demonstrate empathetic social concern; equity centred national development; apply ethical principles; commit to professional ethics & responsibilities and norms of the scientific practice

**Individual and teamwork**

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

**Communication**

Communicate effectively on physical science activities with the scientific community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions

**Project management and Finance**

Demonstrate knowledge and understanding of the scientific, engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects in multidisciplinary environments

**Life-long learning**

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of socio-technological change

## Programme Educational Objectives

- Function as able practitioners and emerge as leaders in their chosen field

- Excel in their profession with social awareness and responsibility
- Interact with their peers in other disciplines at work and contribute effectively to society in building the economic growth of the country
- Develop an interest in consistent upgrading of skills and pursue higher studies in their chosen field
- Build potential to pursue career paths in teaching and research

### Programme Specific Outcome

- Enhance basic concepts of core areas of Physics, especially in Mathematical Physics, Electromagnetism, Classical Mechanics, Statistical Mechanics and Quantum Mechanics for unravelling and comprehending the diverse phenomena observed in nature.
- Execute general Physics and research-oriented experiments with appropriate analysis for proper interpretation of results and to initiate individual projects and present and corroborate research findings.
- Carry out research/investigation independently to solve practical problems and write/present a significant technical report/document.

## M-SC-CHEMISTRY

### About the Programme

The two-year M.Sc. Chemistry in collaboration with Mitsui Kinzoku Components India (MKCI) offered by VIT-AP University provides students ample opportunities to choose courses from a broad spectrum of cutting-edge areas in addition to fundamentals of organic, inorganic, physical, analytical and computational chemistry. Sessions on improving technical skills, scientific writing, oral presentations and research expertise have been incorporated in the curriculum. Furthermore, the students will have the scope to pursue their Master's thesis work at various national and international institutions and industries. Overall, through an approach of integrating the in-depth core and elective courses, and research work, instructed and guided by highly qualified and well experienced faculty members, students will gain the foundation and skill in advancing their career in higher studies as well as in industry.

### Key Features

1. SAS inks Industrial MoU with Mitsui Kinzoku Components India Private Limited (MKCI)

2. SAS inks Academic & Research MoU with CSIR-IICT

3. SAS inks Academic & Research MoU with Slovak University of Agriculture in Nitra, Slovakia, Europe

4. Ongoing more than 7 External Funded Projects by DST, SERB, ICMR, UGC-DAE, etc.

5. 14 Indian Patents publications by faculty and students of SAS

6. 123 research publications by faculty and students of SAS

7. UG/PG students are working on many innovative projects/research

8. Reputed faculty members with Ph.D./Postdoctoral qualification

## Career Prospects

The M.Sc. Chemistry in collaboration with Mitsui Kinzoku Components India (MKCI) graduates from VIT-AP University will be highly eligible for employment in a wide variety of industries such as pharmaceuticals, petrochemicals, polymers, food processing etc. The students will be qualified to appear for various positions in government organizations and PSUs including ONGC, HPCL, IOCL, BPCL, NTPC, GAIL, etc. Moreover, the opportunity to pursue higher academic degrees from reputed Indian and foreign universities will be widely opened to the students.

## Program Outcomes

After successful completion of the program a student is expected to have abilities to

**Disciplinary Knowledge**

Apply advanced concepts of organic, analytical, physical and inorganic chemistry to solve complex problems to improve human life.

**Problem analysis**

Identify, formulate, research literature, and exhibit critical thinking skills to analyse complex data driven problems, reaching substantiated conclusions using the fundamental principles of Chemical sciences.

**Design/Development of solutions/Problem Solving**

Design experiments, analyse, synthesize and interpret data to provide solutions to different industrial problems by working in the pure, inter and multi-disciplinary areas of chemical sciences.

**Conduct investigations of complex problems**

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

**Modern tool usage**

Create, select, and apply appropriate techniques, resources, and computing tools including prediction and modelling to complex decision- making activities related to Chemistry with an understanding of the limitations.

**Chemistry and society**

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

**Environment and sustainability**

Understand the impact of the professional chemical science solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**Ethics**

Demonstrate empathetic social concern; equity centered national development; apply ethical principles; commit to professional ethics & responsibilities and norms of the chemical science practice.

**Individual and teamwork**

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**Communication**

Communicate effectively on complex chemical science activities with the scientific community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

**Project management and Finance**

Demonstrate knowledge and understanding of the scientific, engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

**Life-long learning**

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of socio- technological change.

## Programme Educational Objectives

- To excel as specialists and leaders in their chosen field.
- Function in their profession with social awareness and responsibility.
- Interact with their associates in their workplace and society, and contribute to the betterment of the country.
- To be successful in pursuing higher studies in their chosen field and pursue career paths in teaching or research.
- Develop a sense of social responsibility, ethics, and equity to transform themselves into commitment-oriented professionals with a strong attitude towards sustainable development of the nation.

## Programme Specific Outcome

- Develop analytical skills and problem-solving skills requiring application of chemical principles. Apply appropriate techniques for the qualitative and quantitative analysis of chemicals in laboratories and in industries.
- Acquire capabilities for independently carrying out research/investigation to solve practical problems and write/present a substantial technical report/document.
- Have a sound knowledge about the fundamentals and applications of chemical and scientific theories.
- Learn about the potential uses of analytical industrial chemistry, medicinal chemistry and green chemistry.

# M-SC-DATA-SCIENCE

## About the Programme

Many industry experts and popular business publications have attested that ‘data' in the 21st Century is, like ‘oil' in the 18th Century, the most valuable resource in the world. Hence, the smooth functionality of everything from the government to local companies depends on how we understand data, without which the progress would halt. For those who realize the fundamental value of data and learn to extract and use it, there will be huge rewards in the current data-driven economy. Data Science is a multidisciplinary field that involves the use of scientific methods, processes, and algorithms to extract insights from the large and ever-increasing volumes of data collected and created by today’s organizations. Data Science drives a significant proportion of global economic activity and is influencing scientific progress across numerous fields. Therefore, this programme is specifically designed to meet the needs of the industry in the areas of Data Science.

## Key Features

1. SAS inks Academic & Research MoU with CSIR-IICT
2. SAS inks Industrial MoU with Mitsui Kinzoku Components India Private Limited (MKCI)
3. SAS inks Academic & Research MoU with Slovak University of Agriculture in Nitra, Slovakia, Europe
4. Ongoing more than 7 External Funded Projects by DST, SERB, ICMR, UGC-DAE, etc.
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6. 595 research publications by faculty and students of SAS
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## Career Prospects

According to Harvard Business Review, “Data Scientist is a high-ranking professional with the expertise and curiosity to make discoveries in the world of Big Data”. As the world is increasingly turning to data for decision making, and the amount of the data being generated is growing exponentially, there is a huge demand for Data Scientists not only in established companies but also in start-ups that work on cutting-edge technology-based products, as they rush to hire Data Scientists to tame their business-critical Big Data. In any organization, a data scientist does the model-driven analysis of its data, to help it to improve planning, increase productivity. Data Scientists are needed to work at various levels of business decision-making, such as tactical, operational, and strategic levels, and to share insights with the business. Therefore, Data Scientists are coveted professionals in industries and sectors such as IT industries, research institutions,

Banking and Financial sector, Educational sector, consulting firms, pharmaceutical companies, automobile industries.

## Program Outcomes

After successful completion of the program a student is expected to have abilities to

**Disciplinary Knowledge**

Apply the knowledge of mathematics, Statistics and Computer Science fundamentals, and a Data Science specialization to the solution of complex data-driven problems.

**Problem analysis**

Identify, formulate, research literature, and exhibit critical thinking skills to analyse complex data driven problems, reaching substantiated conclusions using the fundamental principles of mathematics, statistics, computing and data sciences

**Design/Development of solutions/Problem Solving**

Design solutions for data-driven problems and apply lateral thinking and analytical skills to design systems or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

**Conduct investigations of complex problems**

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

**Modern tool usage**

Create, select, and apply appropriate techniques, resources, and computing tools including prediction and modelling to complex decision-making activities related to data with an understanding of the limitations.

**The Data Science and society**

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

**Environment and sustainability**

Understand the impact of the professional data science solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**Ethics**

Demonstrate empathetic social concern; equity centred national development; apply ethical principles; commit to professional ethics & responsibilities and norms of the data science practice.

**Individual and teamwork**

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**Communication**

Communicate effectively on complex data science activities with the scientific community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

**Project management and Finance**

Demonstrate knowledge and understanding of the scientific, engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

**Life-long learning**

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of socio-technological change.

## Programme Educational Objectives

- Practice value-based holistic and comprehensive learning by integrating traditional and innovative learning practices to match the highest quality standards and groom students to be effective leaders in their chosen fields and career.
- Unleash their hidden talents, creative potential, nurture the spirit of critical thinking and to pursue higher education
- Adopt themselves better to the changing global scenario and gain access to versatile career opportunities in multidisciplinary domains.
- Develop a sense of social responsibility, ethics, and equity to transform themselves to a commitment-oriented professionals having a strong attitude towards sustainable development of the nation.
- Recognize the importance and value of mathematical and statistical thinking, training, and approach to problem solving, on a diverse variety of disciplines. recognize and appreciate the connections between theory and applications;

## Programme Specific Outcome

- Demonstrate an understanding of fundamental concepts in Mathematics, Statistics, and Computer Science and usage of various data analysis tools and computing software used in data science problem solving.
- Gain personal and professional competence to be able to collect, process data, model, analyse, design, visualize to address real-world problems, and interpret results using Data Analytics.
- Acquire capabilities to execute research experiments and projects; to design and develop innovative processes, tools, and algorithms; and to build predictive models and machine learning algorithms to analyse large amounts of information from structured and unstructured data to discover trends and patterns.

- Identify and analyse social, legal, and ethical issues in Data Science; provide solutions to particular domain specific problems by having in depth domain knowledge; get exposure to emerging trends and technologies to prepare students for industry and higher studies.

# DOCTURAL

## Ph.D. Program

### Overview

Research is at the heart of SAS's vision towards becoming an internationally recognized centre of knowledge and learning. The spectrum of research activities at SAS is extensive and distinct. Our researchers work to find out the technological solutions to real-world problems and to innovate new technologies to meet the emerging world needs.Currently, the school has over 90 research scholars pursuing a Ph.D. degree. The faculty members have contributed to academic research publications and have successfully published research papers in over 157 journals and several book chapters. SAS encourages our researchers for collaborative research with faculties from foreign universities and to present their findings on international platforms.

The school has about 50 registered faculties under the Research Grant in Engineering, Management, and Science (RGEMS), an initiative to motivate research that can lead to R&D projects. Undergraduate Research (URE) pursuits are encouraged by the school as well. In collaboration with the faculty, undergraduate students pursue active interdisciplinary research to make intellectual or creative contributions towards advancing science.

### Selection Procedure for International/NRI candidates

Foreign nationals / NRI candidates can only register as full-time scholars through separate link:

https://admissions.vit.ac.in/irapplicationpg/

The following are eligible to apply for a Ph.D. programme:

I. Candidates of foreign nationality who hold degrees from Indian Universities seeking admission to research programmes with the necessary clearance from the Government of India (The Ministry of Human Resource Development)

II. Candidates mentioned above who possess a valid visa will be treated on par with Indian nationals for purposes of admission to the Institute.

III. Foreign nationals with foreign degrees who meet the minimum educational requirements as given in Section 3

IV. Foreign nationals whose degrees are equivalent to Indian degrees mentioned in Section 3 in Engineering / Technology / Science and who hold a good academic record

V. International candidates who have a good working knowledge of English.

Please Note: Candidates with valid GRE and TOEFL scores will be given preference. The case of each foreign applicant will be examined and admission will be offered purely on merit. The fee is payable only as USD.

*NRI Guidelines*:- Indian nationals having degrees from Indian universities and whose parents work outside India may apply as NRI candidates. After receiving their application and required documents, they will be interviewed via Skype or other standard meeting platforms web technology. The fee is payable only as USD.

## Major Areas of Research

- **Department of Mathematics:**

  Nonlinear Dynamical System, Modelling of Memory Devices, Integral Transform, Operator Theory, Hydrodynamics Stability, Thermal Convection, Elasto Hydrodynamic, Algebraic Coding Theory, Cryptography, Fractal, MHD Boundary Layers, Perturbation Methods, Stochastic Differential Equation, Approximation Using Linear Positive Operators, Fluid Dynamics, Nonlinear Mathematical Programming Problem, Solute Transport Modelling, Graph b-coloring, Cosmology and more.

- **Department of Physics:**

  Nanomaterials, Transparent Conducting Materials for Optoelectronic Applications, Photovoltaics, Sensors, Composite Materials, Coatings, Condensed Matter and Biological Physics.

- **Department of Chemistry:**

  Materials and Nanoscience, Homogeneous Catalysis, Supramolecular Inorganic Chemistry, Spectroscopy, Coordination Chemistry, Molecular Machines, Organic synthesis among many other areas.